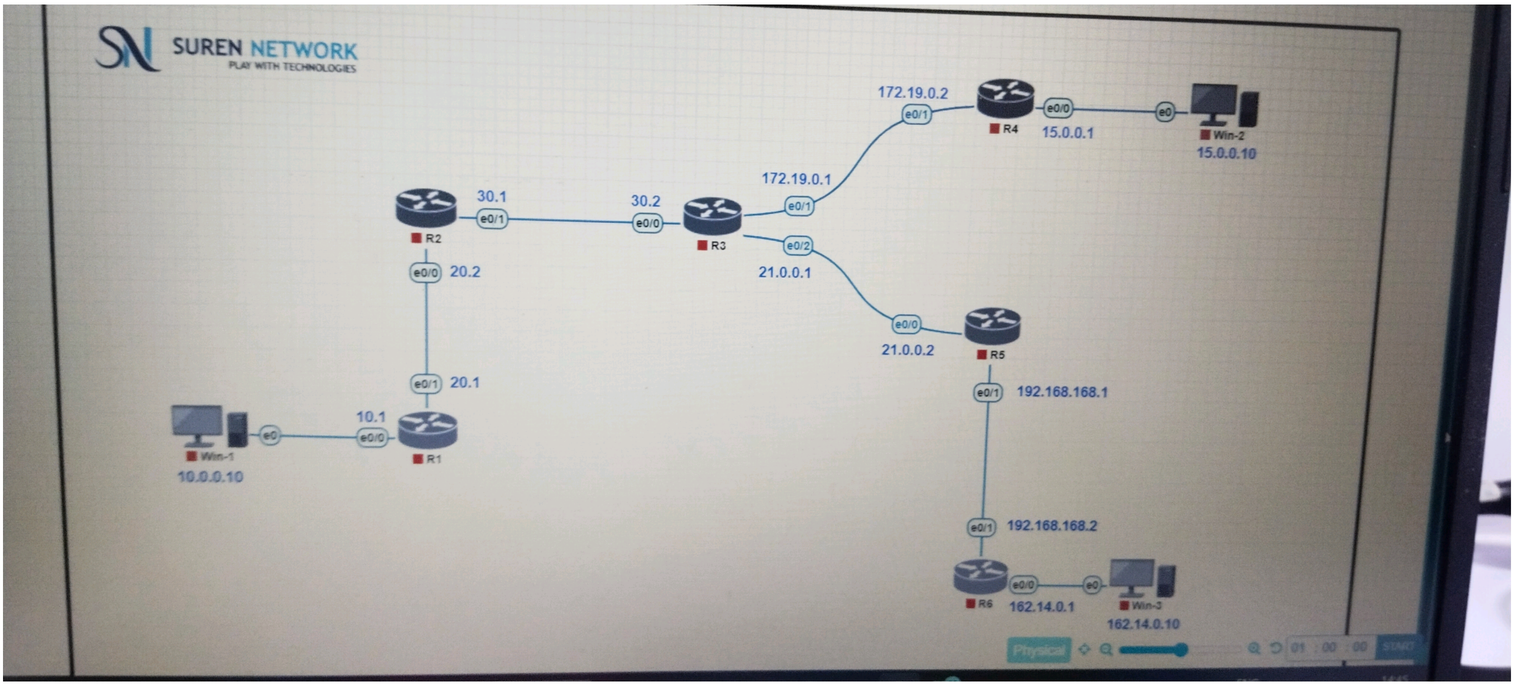
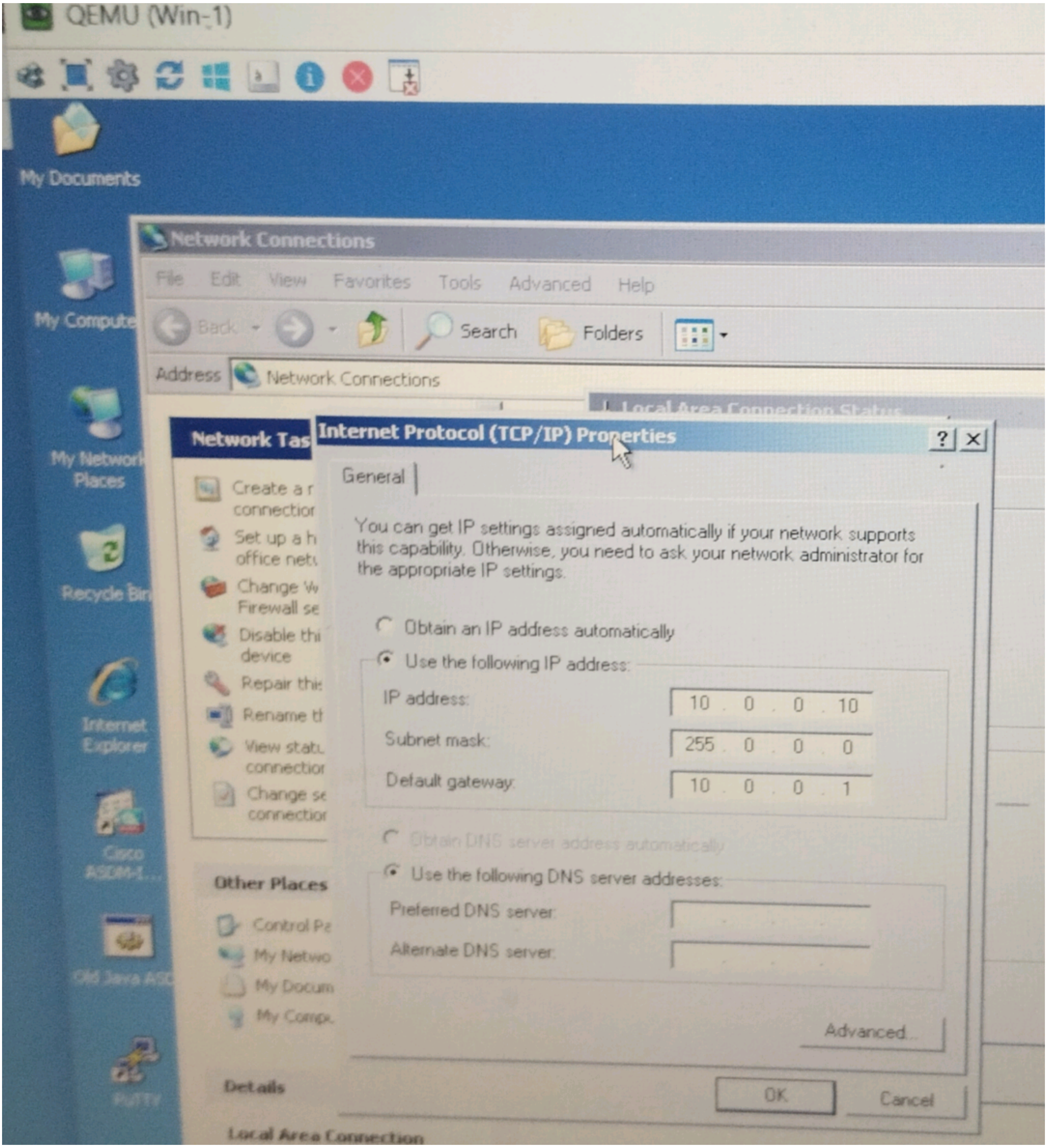
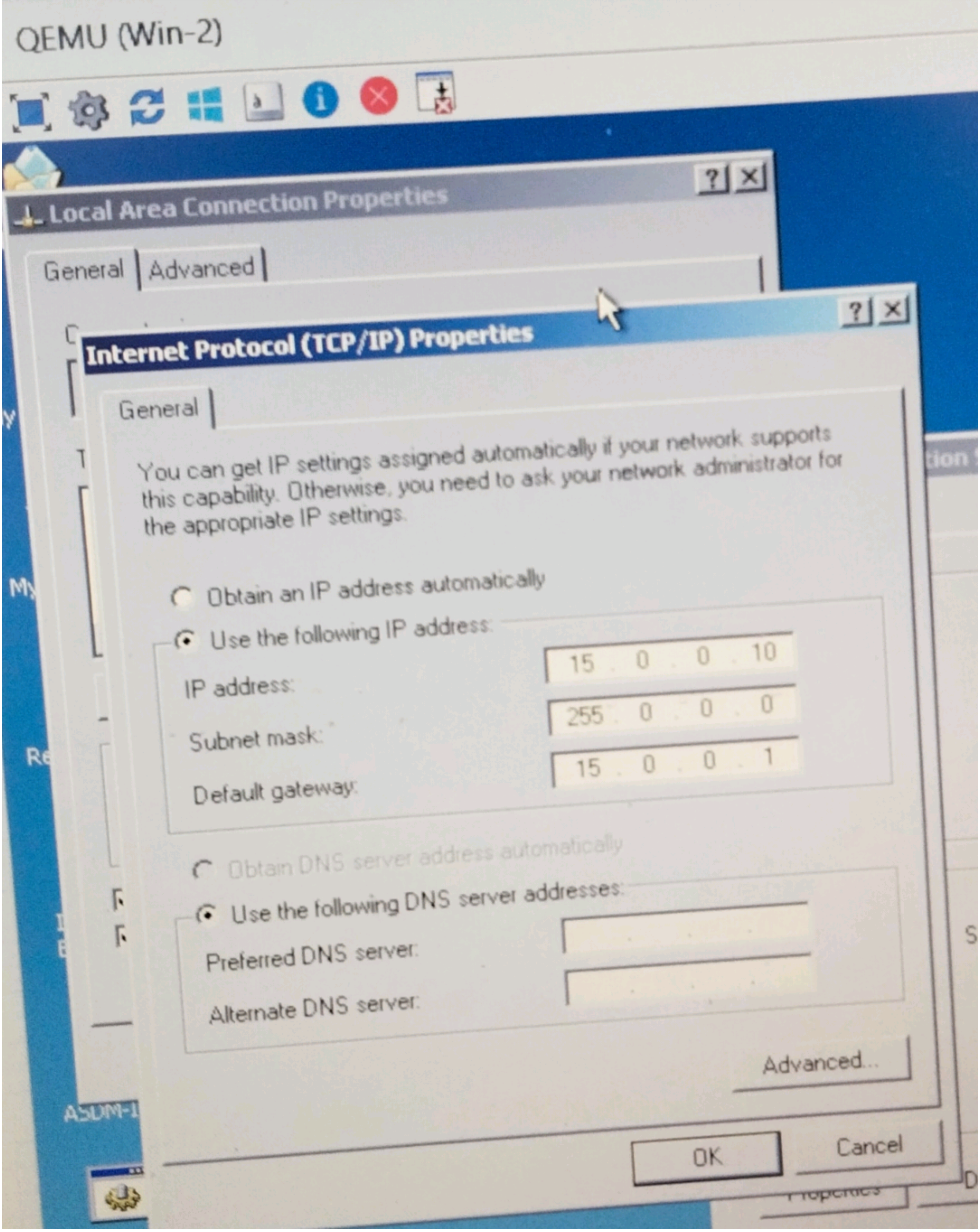
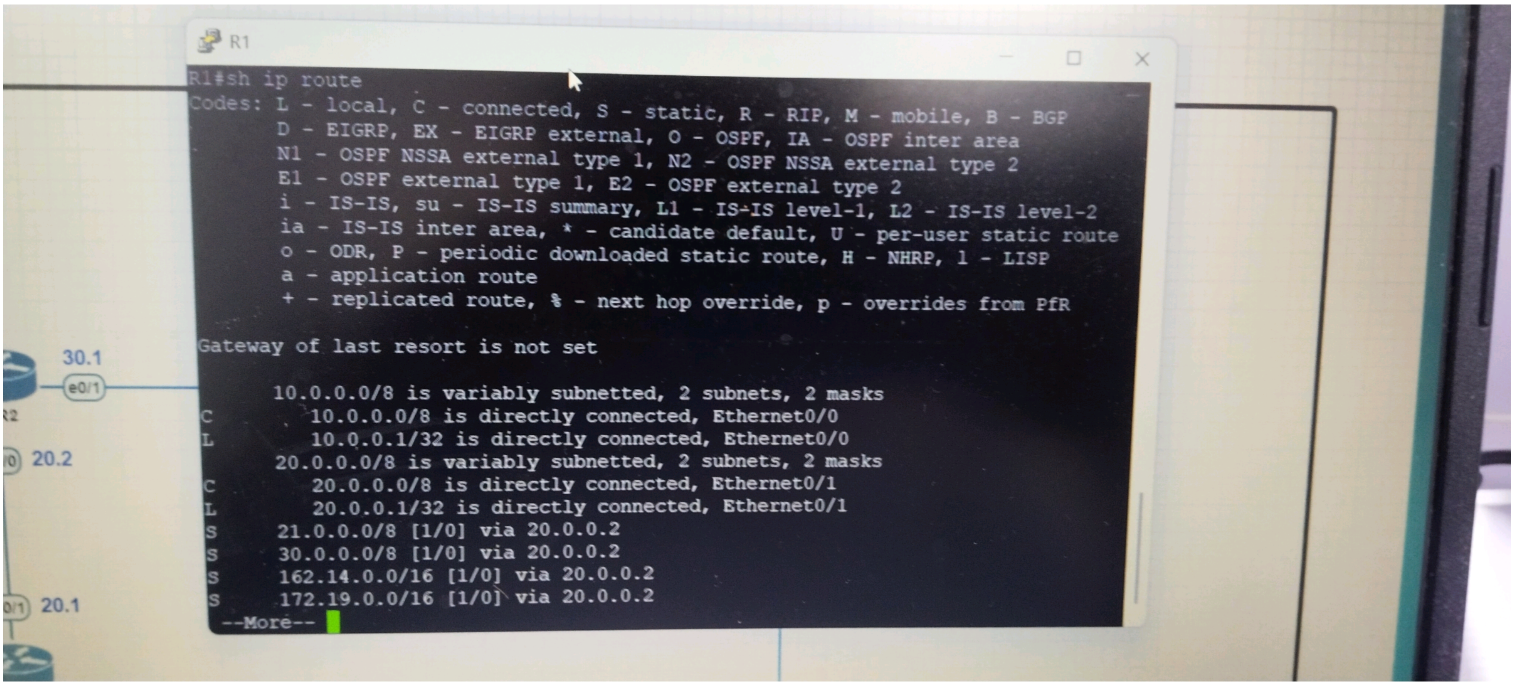

SUREN NETWORK
PLAY WITH TECHNOLOGIES
172.19.0.2
e0/1
R4
e0/0
15.0.0.1
e0
Win-2
15.0.0.10
172.19.0.1
30.1
e0/1
R2
30.2
e0/0
R3
e0/1
e0/2
21.0.0.1
e0/0
20.2
e0/0
21.0.0.2
R5
e0/1
192.168.168.1
e0/1
20.1
10.1
e0
e0/0
R1
Win-1
10.0.0.10
e0/1
192.168.168.2
e0/0
e0
R6
162.14.0.1
Win-3
162.14.0.10
Physical


QEMU (Win-1)
My Documents
My Computer
My Network Places
Recycle Bin
Internet Explorer
Network Connections
File Edit View Favorites Tools Advanced Help
Back
Search
Folders
Address
Network Connections
Network Tasks
Internet Protocol (TCP/IP) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 10 . 0 . 0 . 10
Subnet mask: 255 . 0 . 0 . 0
Default gateway: 10 . 0 . 0 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel
Other Places
Details
Local Area Connection


QEMU (Win-2)
Local Area Connection Properties
General
Advanced
Internet Protocol (TCP/IP) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
15 . 0 . 0 . 10
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
15 . 0 . 0 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Advanced...
OK
Cancel



```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/8 is directly connected, Ethernet0/0
L        10.0.0.1/32 is directly connected, Ethernet0/0
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/1
L        20.0.0.1/32 is directly connected, Ethernet0/1
S     21.0.0.0/8 [1/0] via 20.0.0.2
S     30.0.0.0/8 [1/0] via 20.0.0.2
S     162.14.0.0/16 [1/0] via 20.0.0.2
S     172.19.0.0/16 [1/0] via 20.0.0.2
 --More--
```

R2

```
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#10.0.0.0 255.0.0.0 20.0.0.1
            ^
% Invalid input detected at '^' marker.

R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#exit
R2#
*Aug 25 11:46:15.386: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                20.0.0.2        YES manual up                    up

Ethernet0/1                30.0.0.1        YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 20.0.0.1
S     15.0.0.0/8 [1/0] via 30.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
S     162.14.0.0/16 [1/0] via 30.0.0.2
S     172.19.0.0/16 [1/0] via 30.0.0.2
--More--
```

R3

```
R3(config-if)#ip add 21.0.0.1 255.0.0.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#

R3(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 15.0.0.0 255.0.0.0 172.19.0.2
R3(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.02
R3(config)#exit
R3#
*Aug 25 11:55:49.573: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                30.0.0.2        YES manual up                    up

Ethernet0/1                172.19.0.1      YES manual up                    up

Ethernet0/2                21.0.0.1        YES manual up                    up

Ethernet0/3                unassigned      YES TFTP   down                  down

R3#
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 30.0.0.1
S     15.0.0.0/8 [1/0] via 172.19.0.2
S     20.0.0.0/8 [1/0] via 30.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
S     162.14.0.0/16 [1/0] via 21.0.0.2
 --More--
```

```
R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
R4(config)#exit
R4#s
*Aug 25 12:05:09.887: %SYS-5-CONFIG_I: Configured from console by console
R4#sh ip int br
Interface              IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0            15.0.0.1        YES manual up                    up

Ethernet0/1            172.19.0.2      YES manual up                    up

Ethernet0/2            unassigned      YES TFTP   down                  down

Ethernet0/3            unassigned      YES TFTP   down                  down

R4#
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 172.19.0.1
      15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        15.0.0.0/8 is directly connected, Ethernet0/0
L        15.0.0.1/32 is directly connected, Ethernet0/0
S     20.0.0.0/8 [1/0] via 172.19.0.1
S     21.0.0.0/8 [1/0] via 172.19.0.1
S     30.0.0.0/8 [1/0] via 172.19.0.1
S     162.14.0.0/16 [1/0] via 172.19.0.1
C     172.0.0.0/8 is directly connected, Ethernet0/1
      172.19.0.0/32 is subnetted, 1 subnets
L        172.19.0.2 is directly connected, Ethernet0/1
S     192.168.168.0/24 [1/0] via 172.19.0.1
```

```
R5
R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.0
R5(config)#
R5(config)#exit
R5#
*Aug 25 12:13:59.705: %SYS-5-CONFIG_I: Configured from console by console
R5#sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                21.0.0.2        YES manual up                    up

Ethernet0/1                192.168.168.1   YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R5#
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 21.0.0.1
S     15.0.0.0/8 [1/0] via 21.0.0.1
S     20.0.0.0/8 [1/0] via 21.0.0.1
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/0
L        21.0.0.2/32 is directly connected, Ethernet0/0
S     30.0.0.0/8 [1/0] via 21.0.0.1
S     162.14.0.0/16 [1/0] via 192.168.168.2
S     172.19.0.0/16 [1/0] via 21.0.0.0
C     192.0.0.0/8 is directly connected, Ethernet0/1
      192.168.168.0/32 is subnetted, 1 subnets
L        192.168.168.1 is directly connected, Ethernet0/1
R5#
```

R6

```
R6(config)#sh
% Incomplete command.

R6(config)#exit
R6#
*Aug 25 12:22:47.352: %SYS-5-CONFIG_I: Configured from console by console
R6#sh ip int br
Interface                  IP-Address      OK? Method Status                Prot
ocol
Ethernet0/0                162.14.0.1      YES manual up                    up

Ethernet0/1                192.168.168.2   YES manual up                    up

Ethernet0/2                unassigned      YES TFTP   down                  down

Ethernet0/3                unassigned      YES TFTP   down                  down

R6#
R6#sh int route
              ^
% Invalid input detected at '^' marker.

R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S     10.0.0.0/8 [1/0] via 192.168.168.1
S     15.0.0.0/8 [1/0] via 192.168.168.1
S     20.0.0.0/8 [1/0] via 192.168.168.1
S     21.0.0.0/8 [1/0] via 192.168.168.1
S     30.0.0.0/8 [1/0] via 192.168.168.1
C     162.0.0.0/8 is directly connected, Ethernet0/0
      162.14.0.0/32 is subnetted, 1 subnets
L        162.14.0.1 is directly connected, Ethernet0/0
S     172.19.0.0/16 [1/0] via 192.168.168.1
C     192.0.0.0/8 is directly connected, Ethernet0/1
      192.168.168.0/32 is subnetted, 1 subnets
L        192.168.168.2 is directly connected, Ethernet0/1
R6#
R6#
```

C:\WINDOWS\system32\cmd.exe

```
indows IP Configuration

thernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 162.14.0.10
        Subnet Mask . . . . . . . . . . . : 255.255.0.0
        Default Gateway . . . . . . . . . : 162.14.0.1

C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=1ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 30.0.0.1

Pinging 30.0.0.1 with 32 bytes of data:

Reply from 30.0.0.1: bytes=32 time=3ms TTL=252
Reply from 30.0.0.1: bytes=32 time=1ms TTL=252
Reply from 30.0.0.1: bytes=32 time=2ms TTL=252
Reply from 30.0.0.1: bytes=32 time=1ms TTL=252

Ping statistics for 30.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>
```

ogy

QEMU (Win-2)

:\WINDOWS\system32\cmd.exe

```
ndows IP Configuration

hernet adapter Local Area Connection:

      Connection-specific DNS Suffix  . :
      IP Address. . . . . . . . . . . . : 15.0.0.10
      Subnet Mask . . . . . . . . . . . : 255.0.0.0
      Default Gateway . . . . . . . . . : 15.0.0.1

\Documents and Settings\user>ping 15.0.0.10

inging 15.0.0.10 with 32 bytes of data:

eply from 15.0.0.10: bytes=32 time<1ms TTL=128
eply from 15.0.0.10: bytes=32 time<1ms TTL=128
eply from 15.0.0.10: bytes=32 time<1ms TTL=128
eply from 15.0.0.10: bytes=32 time<1ms TTL=128

ing statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
pproximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

:\Documents and Settings\user>ping 162.14.0.10

inging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>
```

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

```
icrosoft Windows XP [Version 5.1.2600]
C) Copyright 1985-2001 Microsoft Corp.

:\Documents and Settings\user>ipconfig

indows IP Configuration


Ethernet adapter Local Area Connection:

        Connection-specific DNS Suffix  . :
        IP Address. . . . . . . . . . . . : 10.0.0.10
        Subnet Mask . . . . . . . . . . . : 255.0.0.0
        Default Gateway . . . . . . . . . : 10.0.0.1

C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=1ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128
Reply from 10.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```